

ML Engineer with 4 years of taking medical AI from research to product. I built chest X-ray and lung CT finding models, along with the training and evaluation pipelines and internal tools my team reuses.

Experience	ML Engineer	VUNO	Mar 2022 – Present (4 yrs 7 mos)
		Chest X-ray Team Sep 2024 – Present Diagnostic AI products for chest X-ray (VUNO Med®-Chest X-ray™) – Model R&D, ONNX/OpenVINO serialization, and deployment optimization – Built reusable DB-based training and evaluation pipelines with standardized metrics – Developed internal developer tools and established code review practices	
		Lung CT Team Nov 2022 – Aug 2024 Diagnostic AI products for lung CT (VUNO Med®-LungCT AI™) – Built shared 3D lung imaging pipelines to eliminate duplicate code across models – Quantitatively analyzed label errors in external evaluation datasets to guide relabeling by overseas hospitals – Introduced team MongoDB for data integrity and performed on-site hospital validation	
		Digital Oncology Team Mar 2022 – Oct 2022 Digital pathology research and development – 1st place in TIGER Challenge survival analysis track (inference pipeline & TIL analysis) ^[1] – Built containerized pipeline connecting multi-stage models and released open-source – Designed modular reusable library for whole slide image (WSI) processing	
	Researcher	HVCL @ Korea Univ.	Jun 2019 – Feb 2022
		Interactive Segmentation Research – Deep learning models and interactive tools to reduce physician annotation effort ^[C1]	
		Organized International Medical Data Challenges (2 Editions) – Web tool design, implementation, and qualitative evaluation for 3 hospitals ^[2] – Built and operated a digital platform for sharing large-scale pathology datasets ^[3]	
Skills	ML & Serving	Python, PyTorch, ONNX, OpenVINO	
	Data & MLOps	MongoDB, Airflow, MLflow, W&B	
	DevOps & CI/CD	Docker, Git, GitHub Actions	
	AI Tools & Agents	Claude Code, Codex, Gemini	
Education	M.E. in CS	Korea University	2020 – 2022

Awards	1st Place, International Pathology Image Challenge (TIGER Challenge) Radboud University	2022
	1st Place, Korea University Startup Competition Seoul Metropolitan Government, Seongbuk-gu, SH Corporation, Korea University	2021
	1st Place, Dental Panoramic X-ray AI Challenge Healthhub Datathon Track B, Seoul National University Dental Hospital	2020

Publications JOURNALS

[J1]	Analysis of computational tumor-infiltrating lymphocytes in breast cancer from the results of the TIGER challenge TIGER Challenge Consortium Nature Communications	2025
[J2]	PAIP 2020: Microsatellite instability prediction in colorectal cancer Kyungmo Kim*, Kyoungbun Lee*, <u>Sungduk Cho*</u> , Dong Un Kang, Seongkeun Park, et al. Medical Image Analysis	2023
[J3]	Development and operation of a digital platform for sharing pathology image data Yunsook Kang, Yoo Jung Kim, Seongkeun Park, Gun Ro, Choyeon Hong, Hyungjoon Jang, <u>Sungduk Cho</u> , et al. BMC Medical Informatics and Decision Making	2021

CONFERENCES

[C1]	DeepScribble: Interactive Pathology Image Segmentation Using Deep Neural Networks with Scribbles <u>Sungduk Cho</u> , Hyungjoon Jang, Jing Wei Tan, Won-Ki Jeong IEEE International Symposium on Biomedical Imaging (ISBI)	2021
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PREPRINT

[P1]	Joint-Embedding Predictive Architecture for Self-Supervised Learning of Mask Classification Architecture Dong-Hee Kim, <u>Sungduk Cho</u> , Hyeonwoo Cho, Chanmin Park, Jinyoung Kim, Won Hwa Kim arXiv	2024
	* Equal contribution	

Languages	English	OPIc Advanced Low	Jul 2025
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